SDLC PHASE 1 SPECIFICATION

Requirements Analysis & Product Planning

Functional problem-to-feature mapping, technical feasibility, and logic engine specifications for a self-hosted single-user nutrition app.

1. Scope & Operating Principles

The objective of Phase 1 is to define the exact product requirements necessary to build a complete replacement for paid commercial tracking apps. Operating as a single-user monolithic application, all features execute on-device using local SQLite storage, free external APIs (USDA, Open Food Facts), and native Android capabilities.

2. Problem-to-Feature Traceability Matrix

#	Identified Problem	Feasibility & Technical Basis	Proposed Feature Implementation
1	MyFitnessPal Subscription Costs	**100% Feasible:** Local persistence + free APIs eliminate server/sub fees.	**Zero-Cost Full Access:** Unlocked calorie, macro, and log tracking without paywalls.
2	Ad Interruptions	**100% Feasible:** Single-user self-hosted build requires no ad SDKs.	**Ad-Free Minimal UI:** Clean interface with zero advertising modules or external trackers.
3	Weight Tracking Behind Paywall	**100% Feasible:** Stored locally in SQLite `weight_logs` table.	**Native Weight Tracker:** Unlimited weight logging with historical trend visualization.
4	UI Clutter & Slow Navigation	**100% Feasible:** Built with React Native Paper for lightweight UX.	**Streamlined Dashboard:** High-contrast, single-tap food logging with zero newsfeeds.
5	Searching Local/Global Foods	**100% Feasible:** Query USDA (ingredients) + FatSecret (brands).	**Universal Search Engine:** Multi-source API search querying local DB, USDA, and FatSecret.
6	Barcode Scanning Packaged Foods	**100% Feasible:** `expo-camera` paired with Open Food Facts API.	**Live UPC Barcode Scanner:** Camera barcode parser fetching global product macro specs.
7	Meal Tracking Reminders	**100% Feasible:** Triggered locally via `expo-notifications`.	**Scheduled Meal Notifications:** Customizable local push alerts for Breakfast, Lunch, Dinner.
8	Over-Calorie Visual Alert	**100% Feasible:** Triggered when `total_logged > daily_target`.	**Threshold Warning Modal:** High-priority alert modal popping up upon limit breach.
9	Auto-Adjust Calorie Goals	**100% Feasible:** On-device Mifflin-St Jeor TDEE calculations.	**Dynamic Goal Engine:** Auto-calculates Deficit (-20%), Maintenance, or Surplus (+15%).
10	Energy Balance & Fat Loss View	**100% Feasible:** Daily deficit converted to estimated fat burn (3500 kcal/lb).	**Fat Burn Analytics:** Dashboard displaying daily net energy balance and fat mass change.
11	Weekly Weigh-In & Motivation	**100% Feasible:** Scheduled weekly check comparing historical entries.	**Adaptive Feedback System:** Automated weekly prompts delivering trend-based feedback.
12	Profile Edit & Photo Upload	**100% Feasible:** Local files stored via `expo-file-system`.	**Profile Manager:** Local user settings page supporting profile picture uploads.
13	Strava Activity Sync	**100% Feasible:** Strava REST API (`GET /athlete/activities`).	**Strava Activity Engine:** Auto-syncs workout calories to compare Net Intake vs Maintenance.

3. Key Logic Engine Specifications

Mifflin-St Jeor TDEE & Dynamic Adjustment Engine

Calculates daily energy expenditure dynamically without remote backend calls:

- **BMR (Men):** `(10 × weight_kg) + (6.25 × height_cm) - (5 × age) + 5`
- **BMR (Women):** `(10 × weight_kg) + (6.25 × height_cm) - (5 × age) - 161`
- **TDEE:** `BMR × Activity Multiplier (1.2 to 1.725)`
- **Target Goal Adjustment:** Deficit = `TDEE × 0.80` | Surplus = `TDEE × 1.15`

Strava Net Energy & Fat Burn Calculation

Synchronizes external workouts and computes net balance on-device:

- **Net Intake:** `Total Logged Food Calories - Strava Workout Calories`
- **Daily Deficit / Surplus:** `Maintenance TDEE - Net Intake`
- **Fat Mass Burned (Est.):** `Daily Deficit / 3,500 kcal` (1 lb fat ≈ 3,500 kcal)